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ECE 375 hw2

1) Give the microoperations required to fetch and execute ld r3, x+. Only AC and PC can increment itself. MDR is 8 bits wide. AC, PC, IR, and MAR are all 16 bit wide. Internal data bus is 16 bit wide.

   Fetch cycle: This is a 16 bit instruction and occupies two consecutive bytes, therefore two memory accesses are needed to fetch the instruction.

   Fetch
   1. MAR ←PC
   2. MDR←M(MAR),PC←PC+1
   3. IR(15..8) ←MDR
   4. MAR ←PC
   5. MDR←M(MAR),PC←PC+1
   6. IR(7..0) ←MDR

   Execute

   1. MAR(15..8) ←R27
   2. MAR(7..0) ←r26
   3. MDR←M(MAR)
   4. R3←MDR
   5. AC←MAR
   6. AC←AC+1
   7. R27←AC(15..8)
   8. R26←AC(7..0)

2) Give the microoperations required to fetch and execute ICALL. Only AC and PC can increment itself. MDR is 8 bits wide. AC, PC, IR, and MAR are all 16 bit wide. Internal data bus is 16 bit wide.

Fetch
1. MAR ←PC
2. MDR←M(MAR),PC←PC+1
3. IR(15..8) ←MDR
4. MAR ←PC
5. MDR←M(MAR),PC←PC+1
6. IR(7..0) ←MDR

Execute

1. SP←SP+1
2. MAR ←SP
3. MDR←M(MAR), SP←SP+1
4. MAR←SP
5. PC(15..8) ←MDR, MDR←M(MAR)
6. PC(7..0) ←MDR

3 Assembly Code for lab 5 that performs 16bit by 16bit multiplication. Data memory locations $0100 through $0107

0100	0C
0101	00
0102	0F
0103	02
0104	00
0105	00
0106	00
0107	00

3a) X points at 0100 and Y points at 0102. The two values are either 000C and 020F or 0C00 and 0F02. The first multiplication in the inner loop is 0Cx0F that means the two values are 000C and 020F

3b) What are the contents in memory pointed to by LAddrp, LAddrp+1, LAddrp+2 and LAddrp+3 after the loop mul16_Iloop completes for the first time?

This will perform 0Cx0F= 00(rHI) B4(rLO)
And stores into addrp, which starts at 0104.

0104	B4

0105	00
0106	00
0107	00

3C What are the contents in memory pointed to by LAddrp, LAddrp+1, LAddrp+2 and LAddrp+3 after the loop mul16_Iloop completes for the second time?

This will perform 00xOF = 00(rHI) 00(rLO)
And stores into addrp, which starts at 0104.

0104	B4
0105	00
0106	00
0107	00

3D) What are the contents in memory pointed to by LAddrp, LAddrp+1, LAddrp+2 and LAddrp+3 after the loop mul16_Iloop completes for the third time?

This perform 0Cx02 = 00(rHI) 18(rLO)

0104	B4
0105	18
0106	00
0107	00

3E) What are the contents in memory pointed to by LAddrp, LAddrp+1, LAddrp+2 and LAddrp+3 after the loop mul16_Iloop completes for the fourth time?

This performs 00x02 = 00(rHI) 00(rLO)

0104	B4
0105	18
0106	00
0107	00

Final result is 000Cx020F = 18B4

4 Suppose the following array of numbers are stored in data memory. Write a subroutine that (1) determines the largest number among the 8 numbers stored in memory and (2) stores the largest number in memory location $0108.

```
.org 0x0046

Init stack

Rcall max

…

.org 0x0060

.def counter = r15

.def mdr = r16

Max:

        Ldi counter, 8 /Counter for loop

        Ldi x, $0008 /points x to first number

        Ldi y, $0009 /points y to first number

        Ldi mdr, x /using mdr to hold current max, picking first number by default

Loop:

        Ldi x, mpr / reloading mpr with current max

        Sub x, y / subtraction to check which is larger

        Brmi SWAP / if the result is negative then need to swap because Y is larger than
X

        Inc(y) /If we don’t swap then y needs to be pointed to the next number

SWAP:

        Ldi mpr, y+ /loading mpr with new current max

        Cpi counter, 0 /checking if the loop has run 8 times

        Brne loop /jumping back to loop if the loop has not run 8 times.


        Sts $0108, mdr

        Ret
```

5 Determine the location (ie address) and binary code for each instruction in the code developed for problem 4